RESEARCH ARTICLE

Real-World Practice in Secondary Stroke Prevention Following Embolic Stroke of Undetermined Source

Secondary Analysis of the CASPR Registry

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Abstract

Background and Objectives

Multiple clinical trials have failed to show benefit of anticoagulation over antiplatelets in preventing recurrent stroke among patients with embolic stroke of undetermined source (ESUS). However, there are limited data on antithrombotic prescribing patterns and clinical outcomes in a real-world setting.

Methods

This is a subgroup analysis of a multicenter retrospective observational cohort of consecutive adults with ESUS (admitted between 2015 and 2023). Patients were categorized according to antithrombotic initiation by day 7 after stroke or discharge, whichever was sooner: single antiplatelet therapy (SAPT), combination antiplatelet therapy (CAPT) without anticoagulation (AC), or AC with or without antiplatelet therapy (AP) initiated within the first 7 days of index stroke. Differences in treatment strategies were compared between patients admitted between 2015–2018 and 2019–2023 approximating the publication of RESPECT ESUS and NAVIGATE ESUS trials. The primary composite outcome of recurrent ischemic stroke, major bleeding, intracerebral hemorrhage, or death was assessed using unadjusted and adjusted Cox proportional hazards regression.

Results

Of the 2,328 included patients, 1,537 (66.0%) were prescribed SAPT, 561 (24.1%) CAPT, and 230 (9.9%) AC ± AP. Compared with other treatment groups, patients treated with CAPT *were less likely to be female, but more likely to have atherosclerotic risk factors and presented with milder symptoms*. Patients treated with AC ± AP were younger, more likely to have a prior stroke, a higher NIHSS, a large vessel occlusion, and reduced left ventricular ejection fraction. Compared with patients admitted from 2015 to 2018, those admitted between 2019 and 2023 were more frequently prescribed AC ± AP (11.3% vs 5.9%) or CAPT (26.7% vs 17.2%) and less frequently prescribed SAPT (62.0% vs 76.9%, $p < 0.001$). Compared with SAPT, neither CAPT nor AC were associated with any significant difference in the primary composite outcome. Of the secondary outcomes, CAPT was associated with lower risk of death vs SAPT in unadjusted and adjusted models (aHR 0.69, 95%CI, 0.48–0.98).

Discussion

In this large multicenter cohort study, there was an increased frequency of discharge on AC in ESUS patients after the publication of ESUS randomized trials. Treatment with CAPT or AC ± AP was not associated with a reduced risk of recurrent stroke compared with SAPT.

Trial Registration Information

The study is registered on ClinicalTrials.gov (NCT06398366).

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Introduction

Embolic stroke of undetermined source (ESUS) criteria are met in up to 25% of acute ischemic stroke patients and up to 42% in patients aged 18–55 years.[1,2] Multiple randomized clinical trials failed to show a benefit of anticoagulation therapy compared with antiplatelet therapy in the secondary prevention of stroke in ESUS,[3-6] which suggests a potentially greater prevalence and impact of atherosclerotic mechanisms of stroke (as opposed to cardioembolic sources) in this population, for whom anticoagulation would be of limited benefit. Combination antiplatelet therapy (CAPT) has been shown in multiple trials to be superior to single antiplatelet therapy (SAPT) for preventing recurrent atherothrombotic events in patients with recently symptomatic carotid atherosclerotic disease,[7,8] and in patients with noncardioembolic (likely atherothrombotic) stroke mechanisms.[9-11] Considering these data, and considering randomized clinical trial data indicating no superiority of anticoagulation over SAPT in ESUS patients,[5,6] this prespecified analysis of the Cardiac Abnormalities in Stroke Prevention and risk of Recurrence (CASPR) study sought to explore 2 gaps in knowledge. First, we evaluated real-world outcomes in ESUS patients according to antithrombotic treatment (SAPT vs CAPT vs anticoagulation [AC]). Second, we explored how paradigms in treatment changed following the publication of 2 seminal trials confirming no treatment advantage for anticoagulation for all-comers with ESUS.

Methods

The results of this analysis are reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.[12]

CASPR is a multicenter (n = 27 academic Comprehensive Stroke Centers), retrospective observational cohort study conducted to evaluate consecutively admitted patients with acute ischemic stroke (AIS) of undetermined cause despite comprehensive diagnostic evaluation. The methodology of the CASPR study has previously been published.[13] Patients in this analysis were categorized according to antithrombotic initiation by day 7 after stroke (or discharge, whichever was sooner): SAPT without AC, CAPT without AC, or AC with or without antiplatelet therapy (AP).

Definitions

For the purposes of the primary analysis in CASPR, the following definitions were applied to the source data captured by investigators. Left atrial enlargement was classified as severe if the left atrial volume index was >48 cm/m2 (primary definition)[14]; if the left atrial diameter was >5.1 cm for men or >4.6 cm for women (secondary definition, to be used if left atrial volume index was unavailable) or if the left atrial size was qualitatively documented as having "severe" enlargement based on the echocardiography report (tertiary definition, to be used if the aforementioned quantitative findings were unavailable). Patients were included in CASPR irrespective of presence or absence of atrial cardiopathy according to this structural definition. Left ventricular injury was a composite of either reduced left ventricular ejection fraction (20%–40%) among patients without missing ejection fraction data and/or left ventricular hypokinesis/akinesis. Patients could be classified as having cancer in their medical history, but all cancer diagnoses were adjudicated by the data coordinating center. In addition to the specific cancer type (e.g., squamous cell carcinoma of the skin), cancers were identified as locally invasive or metastatic in the data collection instrument. Owing to the heightened risk of thrombotic events in patients with adenocarcinomas, patients with locally invasive squamous, basal cell, and unspecified carcinomas of the skin were adjudicated as not having cancer. This adjudication was conducted before statistical analyses.

Patients were diagnosed with ESUS according to the original 2014 criteria proposed by Hart et al.,[1] with slight modifications to permit evaluation of treatment effect heterogeneity across patient subgroups. The operational definition of ESUS in CASPR required an embolic infarct pattern (cortical, midbrain, or cerebellar infarcts; subcortical ≥2 cm in diameter on diffusion-weighted imaging or ≥1.5 cm on computed tomography; or multiterritorial). Patients with ESUS must have undergone intracranial and extracranial vessel imaging with luminal stenosis <50% in an artery proximal to the region of infarction, transthoracic echocardiography without an intracardiac thrombus, electrocardiography, and minimum of 24 hours of inpatient telemetry without atrial fibrillation. Notably, patients in CASPR were eligible for inclusion if the left ventricular ejection fraction (LVEF) was ≥20% (Hart et al.[1] required ≥30%), if there was an intracardiac tumor or valvular vegetations/lesions (all of which would not have qualified according to the Hart et al. report[1]). Patients with infective endocarditis were originally eligible for inclusion to permit comparative analyses between other valvular pathologies; however, these patients were subsequently excluded owing to underreporting of cases (<0.3% of cohort). We sought to include patients with a lower LVEF than defined by the formal ESUS construct due to the variable antithrombotic practices for reduced LVEF,[15] the inter-rater variability in ventricular function assessments, potential improvement following goal-directed medical therapy,[16] and variability in the method and modality for quantifying ejection fraction, which may differ by up to 20% between methods.[17,18] We also included patients with significant aortic arch atheroma (as defined by the Katz criteria[19]) and those with cardiac valvular disease owing to unclear significance of valvular calcifications, Lambl's excrescence, and other variants which warrant further scrutiny in this type of observational study. Although patients were excluded if they were found to have a new diagnosis of atrial fibrillation (AF) on electrocardiography or telemetry during their hospitalization for AIS, patients were eligible for inclusion if they were later found to have AF on outpatient monitoring. Inclusion of these patients was intended to permit a utilization analysis of various arrhythmia monitoring strategies and outcomes related to such strategies.[20] Patients with other potential embolic mechanisms of stroke such as patent foramen ovale (PFO), nonstenotic carotid plaque, and underlying cancer were also included.

Sites contributed patient data from hospital admissions between 2015 and 2023. Consecutive patients (age ≥18) diagnosed with AIS in the preceding 2 weeks were eligible for inclusion. "MRI-negative" strokes were excluded (owing to potential uncertainty of the diagnosis) with the exception of "MRI-negative" patients treated with thrombolysis or endovascular therapy, who were included. A minimum of 90 days of follow-up data after discharge was an original requirement for inclusion in CASPR, unless the patient experienced a primary outcome event before day 90 (recurrent ischemic stroke, major bleeding event including intracranial hemorrhage, or death). This minimum duration was changed on January 5, 2024, to 30 days due to slow recruitment to permit inclusion of more patients in the primary and secondary analyses.

Outcomes

The prespecified primary outcome was a composite of recurrent stroke (ischemic or hemorrhagic), major bleeding according to the International Society of Thrombosis and Hemostasis,[21] or death due to any cause, with censoring at this event or last known follow-up assessment. Owing to comparisons made across antithrombotic strategies, patients with intercurrent new diagnoses of AF leading to antithrombotic change or PFO with associated closure were censored at these dates of management changes. Any recurrent ischemic stroke events occurring after antithrombotic adjustment for AF or closure of PFO, or bleeding as a consequence of antithrombotic escalation, would have been related to treatment of a defined stroke mechanism (rather than treatment of ESUS). Therefore, treatment and outcomes following antithrombotic changes or PFO closure would not have represented real-world practices or outcomes in a patient with true ESUS. Key secondary outcomes included each of the individual outcomes within the composite end point and intracerebral hemorrhage.

Exposures

The primary exposure was antithrombotic treatment within 7 days or discharge, whichever was sooner. Patients were categorized according to treatment with SAPT, CAPT, or AC with or without AP. AC therapy included warfarin, apixaban, rivaroxaban, dabigatran, heparin or low molecular weight heparin, or bivalirudin. Of the 230 patients treated with anticoagulation (9.9% of cohort), 93 were treated with therapeutic heparin or low molecular weight heparin, 123 were treated with apixaban, 2 with dabigatran, 11 with rivaroxaban, 33 with warfarin, and 1 with bivalirudin (due to heparin-induced thrombocytopenia), with 33 of these patients transitioning between 2 anticoagulants during the first 7 days after stroke admission. The justification for treatment with AC or for combined antithrombotic therapy and duration/adherence/crossover was not captured.

Covariates

Covariates captured in the CASPR registry were published elsewhere,[13] with detailed definitions provided in the Online Only Supplement.

Statistical Analyses

Descriptive statistics were used to summarize patient characteristics. Non-normally distributed continuous data were compared using the Kruskal-Wallis test, with categorical comparisons made using the χ^2 test or Fisher exact test, where appropriate. Outcome events are reported as unadjusted annualized incidence rates with 95% confidence intervals (CI).

The association between the different antithrombotic strategies and the primary and secondary outcomes was assessed using an unadjusted Cox proportional hazards model. Treatment with SAPT was prespecified as the referent comparator for the other 2 arms: (1) CAPT and (2) anticoagulation (outcome comparisons were not made between AC and CAPT). Year of admission for stroke for each group was evaluated for patients with admission date information. A Poisson model was fitted to estimate the average annualized percent change (AAPC) in each antithrombotic pattern (SAPT, CAPT, AC) across year of admission for included patients. Differences in treatment strategy were also compared between patients admitted between 2015–2018 and 2019–2023 (roughly approximating the publication period for RESPECT ESUS and NAVIGATE ESUS trials).[5,6]

For the primary and secondary outcomes, Cox proportional hazards models were fitted and adjusted for age, sex, race, prestroke modified Rankin Scale (mRS), hypertension, diabetes, dyslipidemia, any atherosclerotic vascular disease (coronary artery disease, peripheral artery disease, carotid artery disease, or high-grade aortic arch plaque), prior cancer diagnosis, National Institutes of Health Stroke Scale (NIHSS) score at the time of index stroke, acute intracranial occlusion, presence of left ventricular injury, severe left atrial enlargement (LAE), and PFO (model A). These covariates were selected a priori given their known, causal association with one or more of the outcome measures within the primary composite outcome, and might also have influenced the antithrombotic treatment decision. A secondary model (model B) was generated which included the aforementioned covariates in addition to magnetic resonance imaging findings for the subgroup who underwent such imaging (>90% of the included cohort): the presence of acute infarcts in multiple arterial territories, and Fazekas grades 2–3 white matter disease on T2-weighted MRI. Given the low overall rate of major bleeding events, to avoid model overfitting, we conducted a sensitivity analysis for this outcome in a Cox model that was adjusted for the HAS-BLED score. All adjusted Cox models accounted for clustering by site, with effect estimates summarized as hazards ratios (HR) and corresponding 95% CI. The proportional hazards assumption was evaluated using log-log plots and Schoenfeld residuals if a significant effect estimate was identified.

Further sensitivity analyses were performed to evaluate for heterogeneity of antithrombotic use for the outcome recurrent ischemic stroke across the following prespecified subgroups: age (<50, 50–74, >/= 75), sex, prior stroke, NIHSS (stratified according to minor stroke criteria justifying short-term CAPT [0–3] vs 4–14 vs >14),[10,22] acute multiterritorial infarcts (for the subgroup with available MRI data), and prior or new diagnosis of atherosclerotic vascular disease (ASCVD). ASCVD was captured as a binary exposure using CASPR source data if any of the following criteria were met: prior history of coronary artery disease (n = 303), peripheral artery disease (n = 33), carotid stenosis (n = 22), grade IV or V aortic arch atheromatous disease on echocardiographic imaging (n = 14), or >/= 3.0 mm in maximal axial plaque thickness of either left or right cervical internal carotid artery on computed tomography angiography (n = 381, as reported by local site investigators).

All tests were performed at the two-sided level with a p value ≤0.05 considered to be statistically significant. Analyses were performed in STATA v18.0 (College Station, TX). No adjustments were made for multiple hypothesis testing as analyses are considered exploratory. There was no imputation of missing data. Comparisons between included and excluded patients from the source data set can be found in the original CASPR report (eTable 2).[13]

Standard Protocol Approvals, Registrations, and Patient Consents

This study was approved by the local institutional review board at each participating center with waiver of informed consent because of the retrospective nature of data collection, with executed data transfer agreements with the coordinating enter under IRB # 22–165. This study is registered on ClinicalTrials.gov (NCT06398366).

Data Availability

Data from CASPR will be made available on reasonable request to the corresponding author after execution of a data sharing agreement with the data coordinating center.

Results

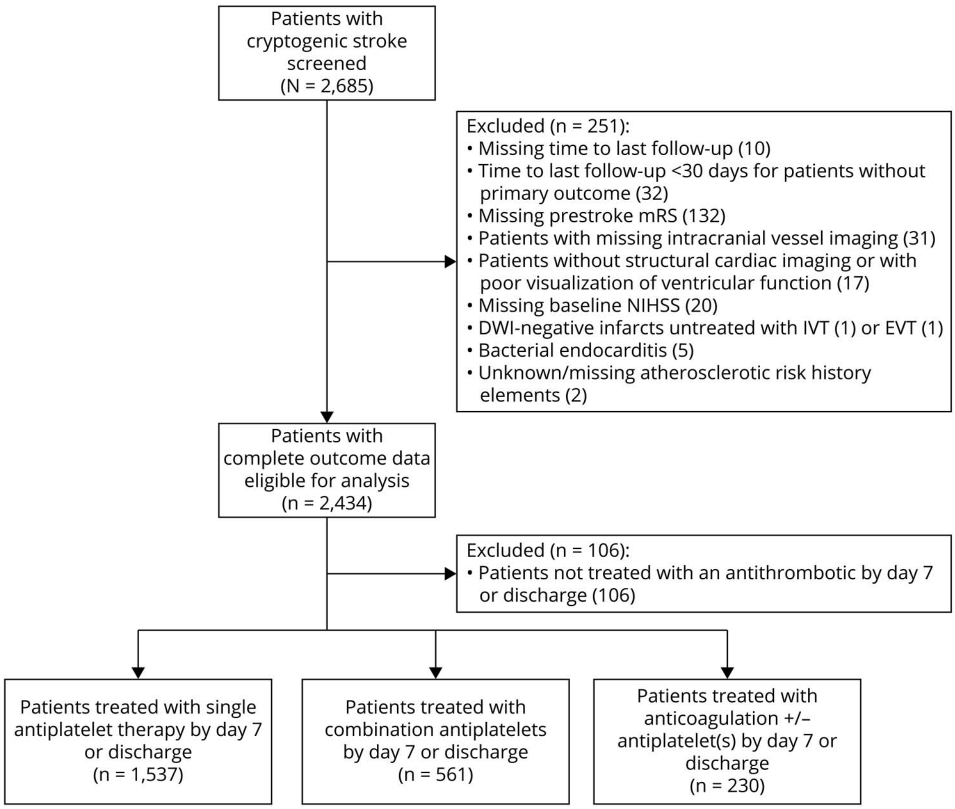
Of the 2,685 patients screened for inclusion, 2,328 patients met inclusion criteria, of whom 1,537 (66.0%) were on SAPT, 561 (24.1%) were on CAPT, and 230 (9.9%) were on AC ± AP (Figure 1). The median duration of follow-up (or to censored event) after stroke was 593 days (IQR 160–1,160), with no differences between antithrombotic groups (Table 1). Of the 230 patients treated with AC, 117 (50.9%) were treated with AC + AP (including 40.0% treated with SAPT and 10.9% with dual AP in addition to AC). Of the 561 treated with CAPT, 515 (91.8%) were on dual AP therapy with the remaining 46 (8.2%) patients prescribed triple AP therapy. Patients treated with CAPT were less likely to be female, and more likely to have a history of atherosclerotic risk factors including hypertension, diabetes, dyslipidemia, coronary artery disease, or any atherosclerotic vascular disease (Table 1). CAPT patients were also more likely to present with milder strokes (CAPT 56.3%, SAPT 38.8%, AC ± AP 35.2%; p < 0.001). Patients treated with AC ± AP were younger, and more likely to have a history of stroke, a higher NIHSS and intracranial occlusion, and more likely to have left ventricular injury. Admission date was captured for 2,218 patients (95.2%), with 19 (70.3%) sites providing data before and after 2019. Compared with patients admitted from 2015 to 2018, those admitted between 2019 and 2023 were as frequently prescribed AC before the index stroke (2.9% vs 1.9%, p = 0.16) but were more frequently prescribed AC ± AP (11.6% vs 5.8%) and CAPT (26.7% vs 16.8%) and less frequently prescribed SAPT (61.7% vs 77.5%, p < 0.01) after stroke. There were declines in SAPT (AAPC -5.6%, 95% CI −7.2% to −4.0%) and corresponding increases in CAPT (AAPC 9.3%, 95% CI 3.8%–14.5%) and AC (AAPC 14.0%, 95% CI 6.5%–20.9%) per year (eTable 1).

Compared with SAPT, all treatment strategies were associated with similar risk of the primary outcome in the unadjusted and adjusted Cox regression models. In the primary adjusted Cox model (Model A), independent predictors of the primary outcome included older age (aHR per year 1.02, 95% CI 1.01–1.03), higher NIHSS (aHR per point 1.03, 95% CI 1.01–1.05), atherosclerotic vascular disease (aHR 1.22, 95% CI 1.02–1.46), and higher mRS (compared with mRS = 0, aHR_3 2.20, 95% CI 1.52–3.19; aHR_4 3.54, 95% CI 2.31–5.41; aHR_5 7.81, 95% CI 2.66–22.91), while the presence of a PFO was protective (aHR 0.61, 95% CI 0.48–0.78).

There was no difference in the secondary outcomes recurrent stroke or intracranial hemorrhage in the adjusted Cox models. However, there was an increased risk of major bleeding with AC vs SAPT in the unadjusted Cox regression (HR 2.04, 95% CI 1.04–3.98, p < 0.05; Table 2). This increased risk persisted but fell out of significance in the primary multivariable regression model (aHR 1.95, 95% CI 0.79–4.81, p = 0.095) and the sensitivity analysis in which the Cox model was adjusted for HAS-BLED score only (aHR 1.96, 95% CI 0.94–4.07, p = 0.068; eTable 2). CAPT was associated with lower risk of death vs SAPT in unadjusted and adjusted Cox models (HR 0.60, 95% CI 0.42–0.85 and HR 0.69, 95% CI 0.48–0.98, p < 0.05, respectively; Table 2). AC was associated with similar risk of death when compared with SAPT in both unadjusted and adjusted Cox models.

Only 12 patients had cerebral microhemorrhages, with 11 ICH events occurring in patients with 0 microhemorrhages

Figure 1 Inclusion Flowchart



DWI = diffusion-weighted imaging; EVT = endovascular therapy; IVT = intravenous thrombolysis; mRS = modified Rankin Scale; NIHSS = National Institutes of Health Stroke Scale.

(0.32%/yr, 95% CI 0.18–0.58%) and 1 ICH event occurring in patients with 1 or more microhemorrhages (0.50%/yr, 95% CI 0.07–3.6%). There was no significant interaction between antithrombotic strategy and microhemorrhage burden in the model (p-interaction = 0.99). The annualized incidence rate of ischemic stroke according to microhemorrhage burden is presented in eTable 3.

Subgroup and sensitivity analyses did not favor any specific regimen in the subgroups except that patients treated with AC ± AP were associated with lower risk of recurrent ischemic stroke in patients without prior history of stroke in the adjusted model (Table 3). In addition, despite significant age differences between treatment regimens, age stratification did not favor a specific treatment in any of the 3 age groups.

Discussion

Since the introduction of the ESUS construct in 2014,[23] 4 randomized trials have shown no antithrombotic treatment regimen to be superior to SAPT for secondary prevention in ESUS patients which may be related to the heterogeneity of patients classified as ESUS.[3-6,23] It is also important to highlight that the lack of benefit of AC over SAPT persisted whether markers of atrial cardiopathy were considered for inclusion[3,4] or not.[5,6] Our study aimed to explore current prescribing practice among providers treating ESUS patients and clinical outcomes, considering CAPT in addition to AC and SAPT.

Our study showed that 1 in 10 ESUS patients during this study period were prescribed anticoagulation with an increase in the proportion of patients treated with AC (and CAPT) after the publication of neutral ESUS trials testing the benefit of anticoagulation over antiplatelet therapy.[5,6] While some of this may be explained by interinstitutional practice differences, the AAPCs for these treatment paradigms were adjusted for intergroup practice differences through clustering. Furthermore, most sites provided patient data from both treatment periods, with only a minority of sites reporting data from only one treatment period. Despite the rise in AC use over time among ESUS patients, we observed no difference in the rate of recurrent ischemic stroke, intracranial hemorrhage, or death with AC ± AP when compared with SAPT. However, the risk of major bleeding was approximately two-fold greater with AC ± AP when compared with SAPT, although this did not achieve statistical significance in multivariable models. These real-world findings are in keeping with previously published clinical trials showing no benefit of AC in ESUS patients when compared with SAPT.[3-6,24]

CAPT was also not associated with significant differences in the rate of recurrent ischemic stroke, intracerebral hemorrhage, or major systemic bleeding when compared treatment with SAPT.

Table 1 Patient Characteristics

	Total cohort (n = 2,328)	Single antiplatelet (n = 1,537)	Combined antiplatelet (n = 561)	Anticoagulation ± antiplatelet (n = 230)	*p* Value[a]
Age[a]	65 (54–75)	64 (53–75)	66 (57–74)	63 (52–73)	<0.01
Prestroke mRS[a]	0 (0–1)	0 (0–1)	0 (0–1)	0 (0–1)	0.09
Female sex	1,157 (49.7%)	796 (51.8%)	237 (42.3%)	124 (53.9%)	<0.01
Race					0.28
White	1,351 (58.0%)	876 (57.0%)	348 (62.0%)	127 (55.2%)	
Black	660 (28.4%)	454 (29.5%)	140 (25.0%)	66 (28.7%)	
Asian	69 (3.0%)	42 (2.7%)	19 (3.4%)	8 (3.5%)	
Other	248 (10.7%)	165 (10.7%)	54 (9.6%)	29 (12.6%)	
Medical history					
Hypertension	1,620 (69.6%)	1,053 (68.5%)	417 (74.3%)	150 (65.2%)	0.01
Diabetes	628 (27.0%)	392 (25.5%)	181 (32.3%)	55 (23.9%)	<0.01
Dyslipidemia	1,091 (46.9%)	678 (44.1%)	307 (54.7%)	106 (46.1%)	<0.01
Prior stroke	366 (15.7%)	199 (13.0%)	109 (19.4%)	58 (25.2%)	<0.01
Coronary artery disease	303 (13.0%)	163 (10.6%)	106 (18.9%)	34 (14.8%)	<0.01
Any atherosclerotic vascular disease	753 (32.4%)	451 (29.3%)	237 (42.3%)	65 (28.3%)	<0.01
Anticoagulation use before stroke	60 (2.6%)	14 (0.9%)	5 (0.9%)	41 (17.8%)	<0.01
History of cancer	255 (11.0%)	165 (10.7%)	58 (10.3%)	32 (13.9%)	0.31
Baseline NIHSS[a]	4 (2–11)	5 (2–12)	3 (1–6)	6 (2–14)	<0.01
NIHSS 0-3	994 (42.7%)	597 (38.8%)	316 (56.3%)	81 (35.2%)	<0.01
LDL[a]	94 (70–120) n = 2,276	95 (70–120) n = 1,505	97 (71–124) n = 547	88 (64–113) n = 224	0.02
A1c[a]	5.8% (5.4–6.3) n = 2,275	5.7% (5.4–6.3) n = 1,503	5.8% (5.4–6.5) n = 547	5.8% (5.3–6.3) n = 225	0.04
Intracranial occlusion	855 (36.7%)	612 (39.8%)	130 (23.2%)	113 (49.1%)	<0.01
Mod-severe LAE	261/2,279 (11.5%)	179/1,502 (11.9%)	59/552 (10.7%)	23/225 (10.2%)	0.62
Left ventricular injury	310 (13.3%)	202 (13.1%)	65 (11.6%)	43 (18.7%)	0.03
PFO	389 (16.7%)	248 (16.1%)	99 (17.7%)	42 (18.3%)	0.57
MRI subgroup					
Multi-territorial infarcts	452/2,102 (21.5%)	276/1,373 (20.1%)	114/527 (21.6%)	62/202 (30.7%)	<0.01
Fazekas grade 2-3	788/2,113 (37.3%)	520/1,378 (37.7%)	203/533 (38.1%)	65/202 (32.2%)	0.28
Days from stroke to last follow-up or censored event[a]	593 (160–1,160)	561 (141–1,183)	641 (202–1,122)	641 (210–1,068)	0.30

Abbreviations: LAE = left atrial enlargement; LDL = low density lipoprotein; mRS = modified Rankin Scale; NIHSS = National Institute of Health Stroke Scale; PFO = Patent foramen ovale.
[a] Median with interquartile range provided.

These real-world data, although limited by retrospective nature and lack of standard duration of antithrombotic treatment, challenge recent evidence suggesting a potential benefit of CAPT over SAPT in high-risk ESUS patients with a higher burden of vascular risk factors.[25] The population from CASPR differs from that of acute secondary prevention trials such as POINT and CHANCE,[10,22] which reported a significant reduction in risk of early recurrent ischemic stroke with CAPT over SAPT. Fewer than half of patients in CASPR had a NIHSS of 0–3, excluding many of the CASPR cohort from eligibility for these trials. However, we observed no antithrombotic treatment effect heterogeneity according to NIHSS strata in this

Table 2 Primary and Secondary Outcomes

	Unadjusted annualized incidence rates (95% CI)			CAPT vs SAPT			AC vs SAPT		
	SAPT	CAPT	AC ± AP	Unadjusted HR (95% CI)	Adjusted HR[a] (95% CI)	Adjusted HR[b] (95% CI)	Unadjusted HR (95% CI)	Adjusted HR[a] (95% CI)	Adjusted HR[b] (95% CI)
Primary composite outcome (recurrent stroke, major bleed, or death)	12.9% (11.7–14.3)	10.6% (8.8–12.7)	15.0% (11.7–19.3)	0.86 (0.69–1.06)	0.90 (0.68–1.19)	0.97 (0.74–1.28)	1.23 (0.94–1.62)	1.06 (0.71–1.57)	1.11 (0.73–1.68)
Secondary outcomes									
Recurrent ischemic stroke	5.9% (5.1–6.9)	6.0% (4.7–7.7)	5.9% (4.0–8.8)	1.11 (0.83–1.49)	1.08 (0.81–1.44)	1.12 (0.83–1.49)	1.12 (0.73–1.72)	1.00 (0.63–1.57)	1.07 (0.66–1.74)
Intracranial hemorrhage	0.5% (0.3–0.8)	0.4% (0.1–1.0)	0.5% (0.1–2.0)	0.83 (0.27–2.56)	0.80 (0.20–3.20)	1.25 (0.37–4.26)	1.12 (0.25–4.96)	0.81 (0.19–3.45)	Inestimable
Major bleeding	1.4% (1.1–1.9)	1.2% (0.7–2.1)	2.7% (1.5–4.9)	0.91 (0.49–1.71)	1.07 (0.47–2.42)	1.15 (0.51–2.60)	2.04 (1.04–3.98)[c]	1.89 (0.89–4.01)	1.95 (0.79–4.81)
Death	6.0% (5.1–6.9)	3.6% (2.6–4.9)	8.1% (5.8–11.4)	0.60 (0.42–0.85)[c]	0.69 (0.48–0.98)[c]	0.75 (0.53–1.07)	1.32 (0.91–1.92)	1.04 (0.66–1.64)	1.07 (0.64–1.77)

Abbreviations: AC = anticoagulation; CAPT = combined antiplatelet therapy; HR = hazard ratio; SAPT = single antiplatelet therapy.
[a] Model A adjusted for age, prestroke mRS, sex, race, hypertension, diabetes, dyslipidemia, atherosclerotic vascular disease, cancer, intracranial occlusion, moderate-severe left atrial enlargement, left ventricular dysfunction, and patent foramen ovale, and clustered by site.
[b] Model B adjusted for covariates in Model A + imaging covariates: acute multiterritorial infarct at index event and grades 2–3 Fazekas white matter disease and clustered by site.
[c] $p < 0.05$.

analysis. Antithrombotic groups in CASPR were also defined by treatment at day 7 or discharge (whichever was later), while POINT and CHANCE defined treatment according to randomization within 24 hours. While the INSPIRES trial extended this window to 72 hours and still observed a benefit with CAPT over SAPT,[26] it is possible patients in CASPR were not initiated on dual AP therapy until well after this window, potentially attenuating any hyperacute benefit. Finally, in CASPR there was a differential distribution of clinical characteristics such as proportionally more Black patients, more patients with cancer, and patients with an acute intracranial occlusion, which may also explain the difference in outcomes between CASPR and other trials of noncardioembolic or cryptogenic stroke patients.

There were notable differences in clinical practice regarding antithrombotic initiation across patient characteristics. For example, patients prescribed AC were more likely to have a history of stroke, cancer, left ventricular injury and present with large vessel occlusion. This could partly be explained by the influence of previously published ESUS trial subgroup analyses on physicians' secondary prevention choices.[24,27,28] In one study evaluating ESUS patients with underlying cancer there was no statistically significant difference between apixaban vs aspirin in risk of major ischemic or hemorrhagic events, but the numerically lower rates of ischemic events favoring apixaban without an increased risk of hemorrhagic complications as compared with aspirin may have influenced the treatment choice.[27] Similarly, AC has been shown to be superior to AP in post hoc analyses of ESUS patients with left ventricular dysfunction.[28,29] In addition, CAPT was more likely to be the initial regimen of choice in patients with mild stroke as well as those with atherosclerotic risk factors and disease which is supported by current AHA/ASA secondary prevention guidelines.[30]

Although these data do not represent randomized patient treatment groups, our findings suggest that the risk of major bleeding with AC or CAPT as opposed to SAPT for the real-world treatment of ESUS is low (but likely higher for patients treated with AC over SAPT). This should be interpreted with caution given the retrospective nature of our study and the risk of selection bias. One study previously showed that despite the addition of AP to AC in patients with AF, there was no benefit and no increased risk of ICH.[31] CAPT was associated with a lower mortality rate when compared with SAPT and warrants further study to determine if a subset of ESUS patients may benefit from this regimen, including those with cardiovascular risk factors, coronary artery disease, and systemic atherosclerotic vascular disease (given these factors were more prevalent in patients placed on CAPT).[32] It is also possible that the lower mortality associated with CAPT may have been secondary to residual confounding by stroke severity, despite attempts made to adjust for this exposure.

While our study offers a large sample with real-world experience across diverse regions and ethnicities, there are limitations primarily related to its retrospective analysis. The justification for different antithrombotic regimens selection was up to the treating physician, and we did not capture the basis for treatment decision. As a multicenter observational

Table 3 Subgroup Analysis for Recurrent Ischemic Stroke

	Unadjusted annualized incidence rates (95% CI)						
	SAPT	CAPT	AC	Adjusted HR[a] (95% CI) CAPT vs SAPT	p_{int}	Adjusted HR[a] (95% CI) AC vs SAPT	p_{int}
Age					0.94		0.14
<50	4.4% (3.1–6.4)	3.9% (1.5–10.4)	6.9% (3.1–15.3)	0.78 (0.32–1.89)		1.27 (0.55–2.93)	
50–74	6.4% (5.2–7.8)	6.4% (4.8–8.6)	5.2% (3.0–9.0)	1.09 (0.71–1.67)		0.75 (0.46–1.21)	
75+	6.8% (5.0–9.1)	6.4% (3.9–10.5)	7.9% (3.3–19.1)	1.02 (0.51–2.04)		1.47 (0.44–4.97)	
Sex					0.99		0.12
Male	5.8% (4.6–7.2)	5.8% (4.2–8.2)	3.2% (1.4–7.2)	1.03 (0.64–1.66)		0.58 (0.27–1.24)	
Female	6.0% (4.9–7.4)	6.3% (4.4–9.0)	8.1% (5.1–12.9)	1.11 (0.76–1.61)		1.35 (0.68–2.69)	
Prior stroke					0.26		0.02
No	5.2% (4.4–6.2)	5.6% (4.2–7.4)	2.9% (1.5–5.6)	1.11 (0.80–1.56)		0.56 (0.34–0.93)[b]	
Yes	10.8% (7.8–14.9)	7.9% (4.8–12.9)	15.7% (9.5–26.1)	0.83 (0.40–1.74)		1.51 (0.76–3.01)	
ASCVD					0.88		0.06
No	5.1% (4.3–6.2)	4.8% (3.4–6.8)	6.8% (4.4–10.5)	1.05 (0.69–1.58)		1.35 (0.76–2.42)	
Yes	8.5% (6.5–11.0)	8.2% (5.8–11.6)	3.6% (1.4–9.7)	1.04 (0.70–1.54)		0.45 (0.19–1.02)	
NIHSS					0.83		0.87
0–3	5.5% (4.4–6.9)	5.6% (4.0–7.8)	4.5% (2.2–9.5)	1.14 (0.78–1.68)		0.69 (0.27–1.80)	
4–14	6.1% (4.8–7.8)	6.6% (4.5–9.8)	7.1% (4.0–12.4)	0.90 (0.53–1.52)		1.02 (0.56–1.83)	
>14	6.5% (4.5–9.5)	7.2% (2.7–19.2)	6.1% (2.5–14.6)	1.24 (0.55–2.78)		1.02 (0.50–2.10)	
Infarct topography					0.64		0.79
Single-territory infarction	5.0% (4.1–6.0)	5.1% (3.7–6.9)	5.1% (3.0–8.7)	1.03 (0.70–1.52)		0.92 (0.52–1.62)	
Multi-territory infarction	8.8% (6.5–11.9)	10.7% (6.8–16.7)	9.8% (5.1–18.8)	1.19 (0.66–2.13)		1.32 (0.60–2.90)	
High grade white matter disease					0.56		0.07
Fazekas grade 0–1	4.2% (3.3–5.3)	4.0% (2.7–5.8)	6.5% (4.0–10.6)	1.11 (0.58–2.13)		1.49 (0.88–2.53)	
Fazekas grade 2–3	8.2% (6.6–10.2)	10.8% (7.8–15.0)	6.0% (2.9–12.6)	1.36 (0.88–2.10)		0.79 (0.34–1.80)	

Abbreviations: AC = anticoagulation; ASCVD = atherosclerotic cardiovascular disease; CAPT = combined antiplatelet therapy; OR = odds ratio; SAPT = single antiplatelet therapy.
[a] Cox proportional hazards model based off model A described in the manuscript, with effects adjusted for age, prestroke mRS, sex, race, hypertension, diabetes, dyslipidemia, atherosclerotic vascular disease, cancer, intracranial occlusion, moderate-severe left atrial enlargement, left ventricular dysfunction, and patent foramen ovale (after excluding the subgroup covariate), and clustered by site.
[b] $p < 0.05$.

cohort, the CASPR study also includes a variation in clinical practices across US Comprehensive Stroke Centers. There was variable pursuit of second tier diagnostic testing for cryptogenic etiologies (e.g., only 35% of included patients underwent transesophageal echocardiography; data not otherwise shown), which may have affected antithrombotic decision making. This will be explored in subsequent analyses. In addition, the duration of treatment, adherence, and crossover were not captured. Despite the comprehensive clinical and diagnostic findings captured in the CASPR data collection instrument, variables including hypercoagulable testing were not collected due to lack of uniformity across participating sites. In addition, no data were captured on covert infarcts limiting the ability to assess their impact in various treatment groups given the recent data suggesting benefit of AC over aspirin when covert infarcts were used as an end point.[33]

In conclusion, our study in a large and heterogeneous group of patients spanning multiple Comprehensive Stroke Centers in the United States showed that 9.9% of patients with ESUS were prescribed AC therapy despite the lack of clinical evidence to support this practice. Furthermore, there has been a rise in AC use since the publication of trials demonstrating no superiority of AC over AP for this condition. However, this practice is likely due to the increasing awareness of other

TAKE-HOME POINTS

- → Despite the absence of evidence from randomized clinical trials favoring anticoagulation over antiplatelets in secondary prevention following embolic stroke of undetermined source, in this real-world multicenter registry there was an increased frequency of discharge on anticoagulation in ESUS patients after the publication of RESPECT ESUS and NAVIGATE ESUS clinical trials.
- → Compared with patients treated with antiplatelet therapy, patients treated with anticoagulation were younger, more likely to have a history of stroke, higher NIHSS, large vessel occlusion, and reduced left ventricular ejection fraction.
- → Neither combination antiplatelet nor anticoagulation regimens were associated with a reduced risk of recurrent stroke, major bleeding, or death when compared with single antiplatelet therapy.
- → Combination antiplatelet therapy was associated with lower mortality when compared with single antiplatelet therapy in this cohort, suggesting that further study is warranted to evaluate this treatment approach.

stroke etiologies such as in patients with atrial cardiopathy in the absence of atrial fibrillation, those with reduced left ventricular ejection fraction as well as patients with cancer, which could potentially benefit from anticoagulation compared with aspirin monotherapy.[24,27-29] There were no clinically meaningful differences with respect to the primary outcome according to antithrombotic treatment (SAPT, CAPT, or AC ± AP). The individual decision to treat with AC in these patients was made at the discretion of the treating clinician and likely varies across institutional practices. These treatment decisions are likely informed by many individual patient factors, including prior antithrombotic use, vascular risk factors, left ventricular injury, and embolic topography. A more detailed phenotypic analysis of ESUS patients in the CASPR study is under way and may provide insights into distinct patient profiles which may benefit from one antithrombotic strategy over another.

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